Open Port Discovery using Nmap

Network Exposure Analysis of a Windows 11 Device

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1. Objective

To discover open ports on a device in my local network using Nmap, identify the services running on them, and understand the security risks of network exposure.

2. Tools Used

- Nmap 7.993 (Zenmap graphical interface) on Windows 11
- Windows Command Prompt (ipconfig) to find the network details

3. Scope

The Wi-Fi network I was connected to uses the subnet mask 255.255.0.0 (/16), which is a large shared network. Scanning other devices on it would need the owner's permission, so I limited the scan to my own laptop (10.0.50.83). This keeps the activity safe and authorized.

Item	Value
Target	10.0.50.83 (my own laptop)
Network	IPv4 10.0.50.83, mask 255.255.0.0, gateway 10.0.30.1
Scan date / time	05-11-2026, 13:22 IST
Scan duration	21.43 seconds

4. Steps Performed

- Ran ipconfig in Command Prompt and noted the IPv4 address and subnet mask.
- Opened Zenmap and entered the target 10.0.50.83.
- Ran the scan with the command: `nmap -T4 -A -v 10.0.50.83`
- The scan includes a TCP SYN (stealth) scan of the 1000 most common ports, service and version detection, OS detection and default scripts.
- Noted the open ports and services, researched them, and saved the scan result.

5. Scan Results

Port	State	Service	Version / Details
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows NetBIOS session service
445/tcp	open	microsoft-ds (SMB)	SMB file and printer sharing; message signing enabled and required (SMB 3.1.1)
3306/tcp	open	mysql	MySQL (reported as "unauthorized")
5357/tcp	open	wsdapi	Web Services for Devices (Windows device discovery)

- Host status: up (latency 0.00063 s). 995 of 1000 scanned ports were closed.
- Operating system detected: Microsoft Windows 11 (24H2 to 25H2). Uptime guess: about 2.1 days.

6. Service Research and Risk Analysis

Port	Service	Purpose and security risk	Risk level
135	MSRPC	Lets Windows programs call services on other machines. It has been used to attack unpatched systems, so it should not be reachable from untrusted networks.	Medium
139	NetBIOS	Old file and printer sharing protocol. Can leak the computer name and share information.	Medium
445	SMB	Windows file and printer sharing. It has been the target of major ransomware attacks such as WannaCry (EternalBlue) on unpatched systems. Message signing is required here, which is a good protection.	Medium
3306	MySQL	A database server is running on this laptop. If reachable from the network it can be attacked by password guessing or used to steal data. Nmap could not authenticate, which is a good sign, but the port is still visible on the network.	Medium to High
5357	WSDAPI	Used by Windows for network device discovery. Low risk, but it reveals that the device is on the network.	Low

7. Recommendations (Mitigation)

- Keep Windows updated and Windows Firewall turned on.
- Set the Wi-Fi network profile to Public, so file sharing and discovery are restricted.
- Turn off file and printer sharing and network discovery if they are not needed (reduces ports 135, 139, 445 and 5357).
- Bind MySQL to localhost only (bind-address = 127.0.0.1), use a strong root password, or stop the service when not in use.
- Keep SMB signing enabled and disable the old SMBv1 protocol.
- Run port scans regularly on my own network to check for unwanted open ports.

8. Conclusion

The scan found 5 open ports on a Windows 11 laptop: 135, 139, 445, 3306 and 5357. Most are normal Windows services, but each open port increases the attack surface. The MySQL port needs the most attention because a database should not be exposed to the network. By limiting services, using the firewall and keeping the system updated, the exposure can be reduced. This exercise showed how Nmap helps in finding exposed services and understanding network security.

Appendix: Scan Output (excerpt)

```
Nmap scan report for 10.0.50.83
Host is up (0.00063s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE       VERSION
135/tcp  open  msrpc         Microsoft Windows RPC
139/tcp  open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp  open  microsoft-ds?
3306/tcp open  mysql         MySQL (unauthorized)
5357/tcp open  wsdapi?
Running: Microsoft Windows 11
Nmap done: 1 IP address (1 host up) scanned in 21.43 seconds
```